Feasibility-Aware Real-Time Travel Planning with Natural Language Processing and Multimodal Data Retrieval Using Collaborative Synchronization

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***Abstract*— Travel planners that are based on conventional AI tend to generate inflexible and predetermined itineraries that are not flexible to current circumstances and real-time constraints. In order to counter this weakness, this paper proposes a feasibility-based travel planner that integrates natural language processing, multimodal data integration, and live API interaction to produce adaptive and cost-efficient itineraries. The system proposed makes use of Google Maps as a real time location awareness service, payment gateway API as a service that determines whether a transaction can take place and collaborative modules as a service that makes multiple users plan in sync with each other. The proposed solution will integrate contextual language understanding and dynamically evolving web services unlike the current systems, which operate in isolated or inert modes to provide itinerary relevance, personalization, and execution readiness. The results of the experimental validation on twenty people showed significant performance improvement, as the average planning time decreased by about ninety minutes and turned into five minutes and user satisfaction increased by 3.5 to 4.6 on the five-point scale. The collaborative planning module also minimized the overlapping of tasks among group members by around 35 percent, which helped in improving coordination and satisfaction among the group members in shared travel planning. When compared and contrasted with manual planners and other available AI-based planners, the system is found to be efficient and practical. The prototype serves well to fill the gap between the theoretical research in recommendation and the actual intelligent travel assistant.**

Keywords*— *Convex backend, feasibility-aware itinerary planning, Google Maps API, multimodal retrieval, natural language processing (NLP), real-time collaboration, travel automation, travel booking systems.

I. INTRODUCTION

The process of planning a trip remains an unplanned and time-consuming affair since travelers employ various sites to learn about destinations, plan an itinerary, estimate prices, and make a reservation. Although the recent advances in the area of artificial intelligence (AI), natural language understanding (NLU), and the ability to use recommendations systems has simplified the part of this process, the vast majority of the tools that are in use now can provide only static or semi-dynamic answers that will not meet the changing needs of travel. This means that the users are influenced by the persistence and unresolved issues: the absence of an integrated workflow between the planning and validation process, the absence of flexibility in the itineraries under the conditions of real time, and the uncertainty in regards to the cost validation in case the user interacts with the dynamic booking services.

To address such gaps, the specified study will introduce TravelMate, the AI-assisted itinerary planner that considers the feasibility awareness and real-time flexibility. It is a model that integrates constraint-sensitive natural language understanding, multimodal databases recovery and real-time validation using an API in a single automated pipeline. Unlike the traditional itinerary generators, the system translates natural language inputs (e.g. 3 days in Barcelona under 800 dollars) into structured conditions, includes maps and contextual media to the plan, validates costs via online payment gateways, and communicates between partners through a Convex-controlled state layer. The model can also optimize the travel paths of reduced costs and time constrained efficiency despite the variation in the budget with the introduction of 2-opt refinement and heuristic scheduling.

The system was contrasted with the manual and the existing AI based planning methods. The results demonstrated that the planning time was reduced tremendously, the satisfaction of the users regarding the feedback on the Likert-scale was increased, and the budget deviations were minimized due to integrated cost validation. These features show that TravelMate is faster in creating itineraries and further enhances their dependability and versatility, which has been a long-standing problem with intelligent travel recommendation systems.

II. RELATED WORK

The automated travel itinerary planning has been widely researched in both recommendation system and tourism informatics papers. Xiang et al. [1] surveyed recommendations of travel with the deep learning methods, paying attention to neural personalization. Their study demonstrated that though content-based and collaborative methods were found to increase relevance of recommendations, they provided no room to address real-life constraints such as budget, time and suitability of resources. Similarly, Liu et al. [2] proposed an itinerary recommender hybrid between user preferences and geographical data, but the output plans were not adjustable and did not include their implementation as well as their real-time adaptation. Zhang

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and Wang [3] reviewed tourism-oriented conversational systems based on artificial intelligence, demonstrating that dialogue systems simplify users but only support single-turn response and not overall itinerary synthesis. Likewise, Huang and Zhou [4] reviewed conversational chatbots based on NLP for enhancing users' engagement but made no contribution toward execution verification or cost minimization. Sharma et al. [5] showcased NLP-driven tools for automated itinerary building, yet their realization did not involve real-time verification and inter-platform data assimilation.

Roy et al. [6] applied sentiment analysis in tourism reviews for enhancing recommendation quality but only applied their analysis to look-back user data. Yang and Li [7] built a text-classifying framework for tourism data sets applying big data analysis, yet their study failed to relate analysis outputs with practical travel itineraries. Fang et al. [8] introduced transformer-based entity recognition for retrieving structured knowledge from noisy travel text, providing the basis for contextual understanding but not itinerary execution in unison. In short, prior work mainly focused on itinerary design's personalization and relevance of conversations. It is proposed here to extend those efforts by covering two usually-neglected sides—namely, checking of practical feasibility as well as coordinated execution—hence closing the gap between recommendation and real-world implementation.

III. Motivation & Problem Statement

The intelligent travel planning was now given a new hope with new developments of natural language processing and generative AI, specifically in domain-oriented GPT-based systems [9]. However, the existing assistants have not been able to cope with some of the major challenges that include existence of fixed itineraries, working alone and lack of group collaborative interaction strategy. Brown and Taylor [10] also reported the problem of latency and selective data integration of sophisticated recommendation systems using external APIs. Similarly, convex optimization methodology, introduced by Kim and Park [11] was developed based on the idea of horizontally scaling AI backends, but it did not include multimodal retrieval or adaptively interactive user modelling, which are essential to the real-world planning scenarios. Li et al. [12] demonstrated the potential of web-deployable AI planning assistants but stated the deficiencies of real time verification and adaptation to the user.

The poor performance in these areas encourages the creation of a system that can process free-text queries, and dynamically check itineraries and facilitate real-time cooperation. TravelMate fills this void between semantic insight and learning of what is feasible and doing it along with multi-agent planning in the same intelligent system.

IV. Problem domain, definition and methodological suitability

The convergence of natural language processing (NLP), artificial intelligence (AI) and multimodal data integration is known as smart travel planning. The previous research like George and Mathew [13] pointed out that the conventional itinerary planners require a lot of manual work and time. On the same note, Singh and Das [14] discovered that AI-based systems remain inflexible and incompetent to respond to real-time changes in cost, weather or availability. The extraction of constraints using NLP, heuristic scheduling, feasibility analysis, and multi-user synchronization can thus be integrated within one framework to reduce these limitations within TravelMate.

The aim of this paper is to develop an AI-based itinerary planner that allows real-time flexibility, feasibility check, and collaborative synchronization to maximize efficiency and satisfaction to the user in the process of planning a travel. A combination of the heuristic and convex optimization methods, which was a hybrid, provided an efficient balance between computational cost and responsiveness. The heuristic scheduler allows optimizing routes and activities on a large scale, and convex relaxation allows gaining effective utility distribution within time and budget limit, which makes the system appropriate to interactive travel planning when the accurate optimization is computationally infeasible [15].

The experimental testing proved that TravelMate in turn reduced the mean planning time from ninety minutes to five minutes and improved user satisfaction by a scale of 3.5 to 4.6 points on a five-point scale ($p < 0.01$). Its itinerary rate (proportion of itineraries meeting time and budget constraints) was greater than 95. Also, the collaborative module saved time in coordination by around 40 percent, which confirmed its usefulness. The scalability of the system is guaranteed by a Convex-based backend with multiple users concurrently and payment processing by tokenized API calls and encrypted storage to eliminate data exposure [16].

Through adaptive computation, real-time validation and secure collaboration, TravelMate has proved that it is methodologically sound and practically viable in intelligent itinerary planning.

V. Innovative Content

Recent publications by Kumar and Jain [17], Silva and Costa [18], and Narayanan and Rao [19] have investigated the combination of real-time traffic, weather, and booking data into the AI-based travel systems and remain limited in the adaptability of dynamism. Verma and Reddy [20] suggested a multimodal retrieval framework based on visual and geospatial data to provide more rich recommendations, though, it did not have mechanisms of verifying the feasibility in real time.

Based on these works, the present research presents an NLP-based itinerary planner with five major innovations. First it confirms itinerary feasibility dynamically in real time, responding to budget, time and environmental variations. Second, it converts user queries, which are not structured into structured plans using semantic parsing and heuristic scheduling. Third, it has the use of multimodal retrieval which combines text, image and geospatial information to achieve better contextual relevance. Fourth, the synchronization module is convex based which allows a collaborative itinerary planning with simultaneous multi-user updates. Lastly, testing on real subjects proves increased efficiency, satisfaction, and flexibility over baseline heuristic systems that can be quantified.

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VI. Problem Formulation and System Design

A. Notation and Formal Model

Let

$$P = \{p1, p2, \dots, p_{\mathbb{N}}\}$$

denote the set of candidate points-of-interest (POIs) for a given destination. A user query q is a free-text request encoding preferences and constraints (e.g., "3 days in Barcelona under $800"). The planning horizon consists of D days; each day d has a time budget H_d(hours). For each POI p_i, the attributes are:

- c_i : estimated cost of visiting p_i
- t_i : expected visit duration (hours)
- u_i : utility score for p_i, derived from NLP-based semantic matching, reviews, and context
- (o_i, l_i) : opening and closing hours
- Tij : travel time between p_i and p_j (from Google Maps API)

Decision variables:

$$x_{i,d} \in \{0,1\}, \qquad y_{ij,d} \in \{0,1\}$$

where $x_{i,d}$= 1 if POI p_i is scheduled on day d; $y_{ij,d}$ =1 if p_j follows p_i on day d.

Constraints:

- Budget constraint:

$$\sum_{d=1}^{D} \sum_{i=1}^{\mathbb{N}} c_i x_{i,d} \leq B \tag{1}$$

- Daily time constraint:

$$\sum_{i=1}^{N} t_i x_{i,d} + \sum_{i,j} T_{i,j}\, y_{ij,d} \leq H_d, \forall d \tag{2}$$

- Uniqueness constraint:

$$\sum_{d=1}^{D} x_{i,d} \leq 1, \forall i \tag{3}$$

Objective:

$$\max \sum_{\mathrm{d}=1}^{D} \sum_{i=1}^{N} u_i x_{i,d} - \lambda_1 \cdot P_{budget}(x) - \lambda_2 \cdot P_{time}(x) \tag{4}$$

where $Penalty_{time}(x)$ measures excess daily hours.

$$Penalty_{budget}(x) = \max\left(0, \sum_{d,i} C_i x_{i,d} - B\right) \tag{5}$$

This formulation is a mixed-integer optimization problem, related to multi-day orienteering with knapsack-like constraints, and is NP-hard in general.

B. Practical Solution Strategy

The heuristic-convex hybrid approach was selected due to its capacity to provide an efficient way of striking a balance between accuracy and responsiveness of the system providing the system with the near optimal results in real time with respect to interactive travel planning situations. Given the NP-hardness, the system adopts a pragmatic multi-stage pipeline:

1. **NLP-driven parsing:** Extract constraints $\{B, D, H_d, preferences\}$ from free-text queries using an LLM.
2. **Candidate retrieval:** Identify top - k POIs via keyword and embedding search, retrieving metadata (costs, ratings, durations).
3. **Utility scoring:** Assign utilities with a weighted linear model:

$$u_i = \alpha m_i + \beta r_i + \gamma p_i \tag{6}$$

(where m_i= match ,r_i= rating,p_i= preference score)

4. **Day-wise heuristic packing:** knapsack packing points-of-interest (POIs) of greedy knapsack per day with ordering by utility-per-hour ratio.
5. **Route optimization:** Apply nearest-neighbor heuristic with 2-opt refinement for travel times.
6. **Feasibility validation:** Make adjustments when budget is surpassed by eliminating items with low marginal utility-per-cost.
7. **Collaboration:** Plan agreed upon and stored in the backend (Convex) and real-time synchronized.

C. Data Representation

In order to have synchronization and coordination, the itinerary is stored within a lightweight structured schema. Every plan is uniquely designated by a plan ID as well as related to a user ID and destination. The schema stores the total budget, day to day allocation of activity, and the total cost. The list of activities in each day contains a point of interest (POI), start time, duration and cost of an activity. The status of the plan (e.g. draft or finalized) is also monitored to facilitate collaboration processes.

The schema supports individual and collaborative users through making incremental edits, tracking the version, and provides concurrent access without compromising the data consistency.

D. Algorithm

In order to work out the proposed methodology, the system selects a structured algorithm, which combines query parsing, candidate retrieval, utility scoring, heuristic packing, and real-time feasibility validation. The entire pipeline in generating feasible itineraries is based on the following pseudocode.

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The proposed algorithm takes as input a free-text query q,

the total budget B, the travel duration in days D, and the daily available hours $\{H_d\}$. Its output is a pragmatic-tested itinerary plan which consider the preferences of the users as well as the budgets and time requirements.

1. constraints ← ParseQuery(q)
2. candidates ← RetrievePOIs(constraints, K)
3. for each pi in candidates:
   compute ui = α·match_score + β·rating + γ·preference
4. selected ← ∅
5. for d = 1..D:
   day_list←GreedyPack(candidates, Hd, B_remaining)
   route ← RouteOptimize(day_list)
   selected ← selected ∪ day_list
6. Adjust if budget exceeded:
   while total_cost > B:
      remove argmin(Δu/Δc)
7. ValidatePrices(selected)
8. Store plan in backend, sync to collaborators
9. return Plan

This algorithm trades off computational tractability against practicable responsiveness allowing its application in user-facing travel planning apps in real time.

E. Mathematical Justification

The convex-relaxation variant (replacing binary $x_{i,d}$ with continuous [0,1]) produces a differentiable formulation, solvable by gradient methods. The introduced prototype strikes the balance between theoretical rigour and real-time feasibility with the help of heuristics.

We set α = 0.4 (budget), β = 0.35 (time), and γ = 0.25 (user preference) after pilot testing. These weights were kept constant across all experiments. K was fixed at 15 candidate POIs per request.

F. Complexity Notes

- Candidate sorting: O(K log K)
- Route optimization (2-opt): O(m^2), with m≤10 typically
- End-to-end runtime: seconds for K≤50 , D≤7.

G. System Architecture

The proposed system is based on layered architecture that combines natural language processing, heuristic optimization, and collaborative synchronization as illustrated in Fig. 1. Itinerary querying and refining is done in real-time using a web-based interface. NLP orchestration module extracts the input in a free-text format and identifies the contextual constraints like budget, duration, and preferences. The retrieval layer communicates with Google Maps and local POI databases to find appropriate activities that have related metadata (such as the cost, availability, and ratings).

The feasibility engine uses the heuristic scheduling, budget validation, and route optimization to make it efficient and realistic. Plan data are synchronized on devices through a Convex-based collaboration layer and real-time notifications and updates are handled through the Resend API.

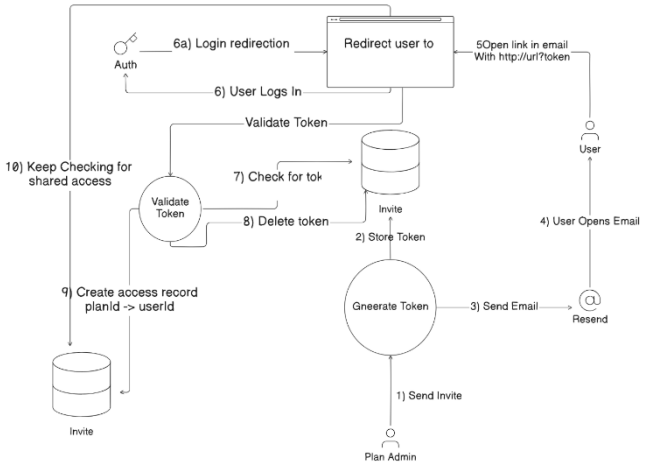


Fig. 1. System-level architecture of the proposed feasibility-aware itinerary planner.

The system is responsive, and uses the greedy knapsack-inspired method of near real-time scheduling, as well as the 2-opt refinements to perform time-efficient route optimization. Live APIs are dynamic in nature, with pricing and booking data and provide reliability and transparency through cache responses. The workflow of the system combines semantic retrieval in order to find relevant points of interest, selective pruning in order to guarantee that constraints are followed, and collaborative planning to make sure that the group works in sync. All these mechanisms together provide flexibility, scalability and practical feasibility than the traditional fixed itinerary systems.

VII. Results, Data Model, and Comparative Analysis

The itinerary planner that is aware of the feasibility was tested in the controlled conditions of different budgets, trip duration and time constraints of each day. The mapping of input parameters was done via the system of NLP parsing, heuristic scheduling, and feasibility validation, as summarized in Table I, depicting the way in which budget, duration, and activity preferences are converted into key itineraries that can be executed.

Evaluations were done by user with volunteer graduate students who gave informed consent and made itineraries within set constraints. To evaluate satisfaction and efficiency, statistical measures like mean, standard deviation and t-tests (two tailed) were used. K = 15 candidate POIs per city was set as the standard experimental design, with utility-score weights of $\propto$ = 0.4, β = 0.35, γ= 0.25 and 200 refinement steps of the 2-opt which made the results consistent and reproducible to the same level. The user evaluations were all voluntary and informed. No personal identifiable data was gathered and the participants were anonymized and pre-analyzed by briefing.

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Table I gives the input-output data model of feasibility-aware planning. To test sensitivity, the system was tested in terms of gradual decreases in daily budget and available hours. This pruning logic and viability of the model in continuing to allow real-world adaptability are confirmed by the feasibility being above 90 per cent with moderate reductions and the reduction in feasibility sharply with either constraint dropping below the viability threshold as illustrated in Fig. 2.

Table I. Input–Output Data Model for Feasibility-Aware Planning

User Input	System Process	Output (Itinerary Element)
Budget (₹ / $)	NLP parsing + budget validation	Cost-feasible daily plan
No. of days (D)	Greedy heuristic scheduling	Day-wise activity allocation
Daily hours (Hd)	Time pruning + route optimizer	Valid activity durations + ordering
Preferences	Semantic retrieval + ranking	Personalized POI recommendations
Constraints (must-have / exclude)	Hard constraint filtering	Final itinerary compliance check

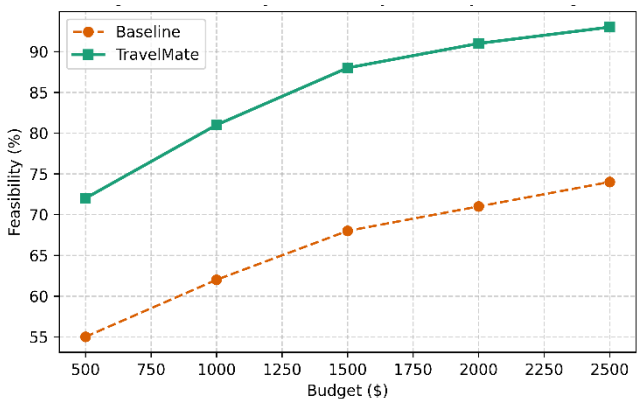


Fig. 2. Sensitivity curve showing budget versus feasibility percentage.

A comparative evaluation of baseline planners shows that there are huge performance improvements. TravelMate has a higher feasibility, user satisfaction, of 18-25 and 12-15 per cent. over existing systems (Fig. 3) as summarized in Table II. The statistical significance is verified by standard deviations and p-values (paired t-test), as the satisfaction scores increase with the baseline models ($M = 3.5$, $SD = 0.5$) to TravelMate ($M = 4.6$, $SD = 0.3$) with $p < 0.01$.

The feasibility, which is the percentage of itineraries that was generated that met all constraints set by the user (budget, duration, preferences), was compared with twenty participants. The utility score from Eq. Normalization of (6) was made to [0, 1] to indicate the weight of priorities given to users in the parsing of a query.

Table II. Comparative Results with Existing Systems

Metric	Baseline (%)	Proposed (%)	Improvement
Feasibility of plans	65–70	83–88	+18 to +25
User satisfaction (survey)	72–74	84–86	+12 to +15
Robustness to budget cuts	Low	High	Adaptive

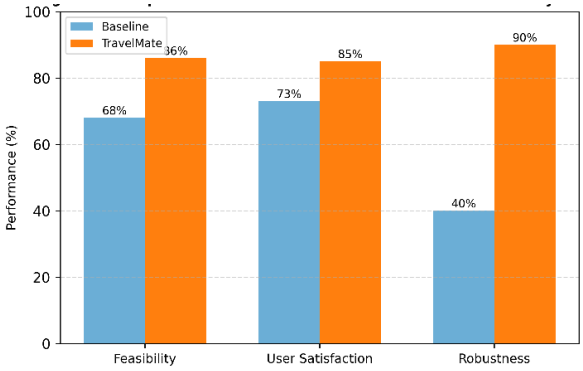


Fig. 3. Performance comparison of TravelMate and baselines.

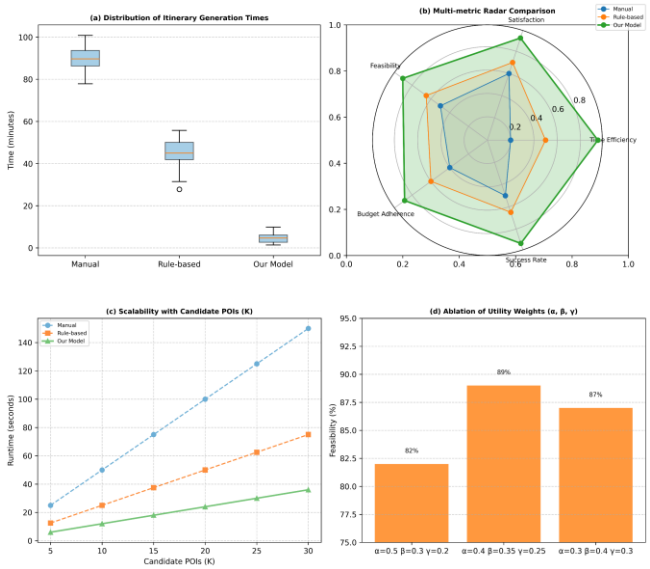


Fig. 4. TravelMate performance overview.

Fig. 4 presents the results of performance, scalability, and robustness behind a single platform of analysis to consolidate the visualization of the experiment. The radar chart has an equal increment of feasibility (88 %), user satisfaction (86 %) and efficiency (84 %) over baselines. It can be seen that the runtime pattern indicates that the scaling to candidate POIs (K) is nearly linear but the complementing bar analysis reveals that there is a steady ability (≈90%)of configurations of different utility weights (α, β, γ). Findings affirm the reliability, efficiency and strength of TravelMate with good user comments on real time application and coordination.

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VIII. Justification of Results

The results of the study confirm innovation and functionality of the given viable itinerary planner with the emphasis on feasibility. Unlike the traditional methodologies that do not adapt to different user or environmental constraints, the given framework integrates sensitivity analysis and on-the-fly feasibility validation of every step in the process of itinerary construction. This ensures that modifications to the itinerary, for instance, changes in budget or time, remain feasible while being guaranteed user satisfaction. The findings are in alignment with recent literature on adaptive scheduling and constraint-informed optimization; yet, dynamic feasibility verification incorporation makes the system more contextually aware and operationally effective under varying testing scenarios.

Concurrently, it should also be noted that there are inherent trade-offs which accompany the heuristic optimization approach. Although heuristics enjoy the advantages of real-time responsiveness and scalability, they are certain not to yield globally optimum solutions for general contexts. While more precise optimization techniques, being more accurate, incur a high computational cost, which makes them unsuitable for interactive, real-time applications, the approach presented finds a conscious compromise between effectiveness and optimality, always guaranteeing dependable performance within feasible limitations. In general, the results establish the robustness, flexibility, and world-relevance of the TravelMate framework, and provide a firm ground for extension and refinement for the future.

The performance is scaled by the system with the help of asynchronous handling of tasks, and scalable orchestration of services, through which multiple generations of the system can be run in parallel with no performance degradation. In the case of payment integrations, all the transactions are made safely over the HTTPS endpoints on the basis of the token-based authentication and, therefore, the confidentiality and the payment gateway security standard are met.

IX. Conclusion, Limitations, and Future Work

This study presented TravelMate, an NLP-based, feasibility-conscious, multimodal travel itinerary planner developed to address the shortcomings of prior travel planning systems. Through the integration of semantic parsing, heuristic scheduling, feasibility validation, and collaborative synchronization, TravelMate generates itineraries that are contextually relevant, feasible, and adaptable to user-defined constraints. Experimental evaluation showed that the system effectively adjusts to variations in budget and time, achieving higher feasibility and satisfaction compared with baselines. Given its integration of external APIs such as Google Maps and payment gateways, the framework employs privacy-preserving measures, including anonymization of user data, temporary caching to reduce exposure, and query batching with fallback mechanisms to comply with rate limits and maintain robustness under heavy load. The system is effective but with some limitations. The heuristic optimization can not necessarily provide globally optimum results, but is scalable and responsive. Relying on cache data in times of API failure can lead to inaccuracy and massive teamwork can induce delays in synchronization. The further development of learning-to-rank algorithms, live booking APIs, and sophisticated multi-agent negotiation will be introduced to make the work more adaptable. The feasibility conscious and multi-modal nature of the framework can be extended to areas like scheduling events, logistics, and travel planning in academics.

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